

Yongjun Rong

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EDUCATION

Jun.,2006, Master of Computer Science, Texas Tech University, Lubbock, TX, USA

Feb.,1996, Master of Engineering, Shanghai JiaoTong University, Shanghai, China

Jul.,1991, Bachelor of Engineering, Changsha Railway University, Changsha, Hunan, China

SUMMARY:

A lead cloud architect engineer with strong hands-on digital transformation experiences in finance industry to migrate on-premises legacy monolithic tightly coupled system to cloud-native Microservice loosely coupled system and refactor strong synchronous ACID relational data consistency system to asynchronous event driven architecture (EDA) eventual consistency system, designed and implemented Internet scale cloud-native and hybrid enterprise system for financial/insurance and tele-communication industries. A full stack lead software engineer with emerging technologies worked as lead architect engineer, lead developer, lead infrastructure/DevOps engineer, senior system/network engineer for Barclays, Oracle, Comcast, IBM and Sun with over 20 years professional experiences focusing on back-end distributed system architecture design and implementation, system/network administration, infrastructure automation/orchestration. A tech lead and player with excellent verbal and written communication skills engaged in small or large team with agile methodology. A hard worker, who will bring valuable experiences, knowledges/skills and expertises to the team's success, is looking for enterprise architect, technical architect, cloud infrastructure architect or tech lead opportunities within Metro Philadelphia area or allowing work remotely.

SKILLS:

- Cloud (AWS/GCP/Azure) infrastructure Design, Implementation and Automation (EC2,EKS/AKS/GKS, ECS, Lambda, S3, RDS, DynamoDB, VPC, API Gateway, ELB/ALB, R53, PrivateLink, IAM, Docker, Terraform, Kubernetes, Serverless, IaC)
- Enterprise Architecture for large Finance/Insurance/Telecommunication organization (Strategic technology Domain, Distributed system, Microservice, EDA, streaming processing, integration, message transportation, architecture handbook, ServiceFirst)
- Digital transformation architecture engineering from on-premises legacy monolithic tightly-coupled system to cloud-native distributed loosely-coupled system (AWS, JEE, SQL, NoSQL, Microservice, CAP, BASE, Clean Architecture)
- Cloud/On-premises infrastructure DevOps/System/Networking administration Automation and Orchestration(AWS, Docker, Kubernetes, Terraform, Chef/Puppet/Ansible, CI/CD, GIT, Maven/Gradle, Linux HA, SAN, Ruby/Python/Bash, Nexus/Artifactory, Jenkins/Bamboo)
- SaaS Enterprise Application (ERP/CRM/Accounting/HR) Integration (EAI) Architecture Design and Implementation (AWS, VPC Endpoints, PrivateLink, SOAP/REST, WSDL/XML/JSON,)
- Internet scale loosely-coupled high available/scalable enterprise Microservice system design, development and deployment (Java, SpringBoot, Istio, Node.js, lambda, Rest/JSON, CQRS, event sourcing, kafka, EDA)

HOBBIES

- Pingpong (ITTF Score around 1200)
- IoT or Robot programming (Have iRobot and MIPOSaur at home. Programming against iRobot create SDK and MIPOSaur SDK.)
- IOS/Andriod Apps Development (Developing a Chinese charater learning App)
- Playing with cutting-edge technologies (Docker, Go, Kafka, Kurbernels and Amazon Cloudformation, JavaScript, AngularJs, ReactiveJs, Node.js)
- IFTTT Home Automation (IFTTT, WeMo, EcoBee, Weather Underground, SaaS, REST, Home Automation)

WORK EXPERIENCE SUMMARY:

Employments	Major Accomplishments & Technologies
Dec.,2021 – Current McKinsey & Company Cherry Hill, NJ (WFH),USA Senior Cloud Engineer II (Associate)	• Migration/Refactoring of Mainframe (IBM MQ) EDA system to AWS by using EventBridge/Lambda/Step Function/CloudWatch/EKS/SQS/SNS...etc. Event Driven Architecture (EDA), Terraform, Microservices, Java/SpringBoot/NodeJs, EventBridge/CloudWatch/Lambda/Step Function, DynamoDB/RDS/Oracle, EKS, Json Schema, JOI, OpenAPI/Swagger,
Dec.,2020 – Dec., 2021 Yotascale Cherry Hill, NJ (WFH),USA Staff Engineer	• Lead Architect and Lead Implementation Engineer for Cloud(AWS/Azure/GCP) Cost Analytics/Optimization /Management platform OpenAPI/Swagger, Microservice, Java/Kotlin/SpringBoot, Kubernetes/EKS/AKS, Istio, Terraform, CQRS, Event Store, Event Driven Architecture (EDA), Kafka/ActiveMQ/IBM MQ, Kafka Stream, Docker, AWS(EC2/IAM/S3/ECS/EKS/ELB/Lambda/RDS/VPC/R53/Secret Manager/AutoScaling/SES..etc), Puppet/Chef/Ansible, Git/Maven/Jenkins, Python/Bash/Go, ELK/AppDynamic/Datadog,
Mar.,2020 – Nov., 2021 Gigster Inc. Cherry Hill, NJ,USA Technical Infrastructure/Devops Architect	• Images manipulation/detection software evaluation system Microservice, NodeJs, Kubernetes/EKS, Terraform, RabbitMQ/ActiveMQ, Docker, AWS(EC2/IAM/S3/EKS/ALB/RDS/VPC/R53/AutoScaling/SES/ECR..etc), Puppet/Chef/Ansible, Git/Maven/Jenkins, JavaScript,
Jun.,2017 – Feb., 2020 Barclays Wilmington, DE,USA VP,Lead Architect Engineer	• Delivery Lead for strategic distributed system technology domain • Group-wide Cloud Adoption for Contact Center, Merchant Services and Fraud Terraform, CQRS, Event Store, Event Driven Architecture (EDA), Kafka/ActiveMQ/IBM MQ, Kafka Stream, Docker, AWS(EC2/IAM/S3/ECS/EKS/ELB/Lambda/RDS/VPC/R53/Secret Manager/AutoScaling/SES..etc), Puppet/Chef/Ansible, Git/Maven/Jenkins/Bamboo, Python/Ruby/Scala/Bash/Go, ELK/AppDynamic, DynamoDB/MongoDB/ElasticCache/GridGain, Linux/Windows/AIX/Solaris, aPaaS/VMWare/Citrix,
May, 2014 – June, 2017 Oracle Redwood, CA (WFH@NJ),USA Lead Software Engineer	• Docker Swarm/Compose/Stack CD/CI Test Environment for Cloud SDK/APIs/Connectors • Cloud Enterprise Applications Integration(EAI) connectivity SDK/API plugin framework for Oracle SOA Suite • Oracle Fusion Applications (FA) Business Event/Message-Driven Connector Framework Java, SOA, Web Services, SOAP/WSDL, REST/JSON/XML, XSchema/XSD, DOM, WSDL4J, SDK, API, MVC, Docker, Cloud, Amazon EC2, NoSQL, GIT, Maven, Junit, SoapUI, IaaS, SaaS, AWS,
Oct. 2010 – Apr. 2014 Comcast T&P Philadelphia, PA,USA Senior Cloud DevOps/Infrastructure Automation Engineer	• AutoBuild System for Build/Release more than 200 Microservice projects with GIT/SVN/Maven • Configuration management and deployment automation with Puppet and Capistrano • Continuous Integration (CI) and Continuous Deployment (CD) with Bamboo and Jenkins • Performance Monitoring System for Jetty, JVM, Linux and Splunk • Secured FTP server with virtual users and admins users with complex access permissions • BOM (Bill Of Materials) Administration and Management Application (BAMA) Enhancement and maintenance • Two Steps bare-metal Linux bootstrap system for HP Servers • Linux HA for Git Repo(Gerrit) and Maven Repo (Artifactory/Nexus) Servers • Automation scripts and tools for DevOps • CoreOS and Docker prototyping and testing • SDLC process management and junior team member training and mentoring DevOps, Automation, Docker, CoreOS, Ruby, Ruby on Rails 3, Bash, SDLC, GIT, Maven, Jira/Confluence, Bamboo/Jekins/Hudson, Nexus/Artifactory, Gerrit, CI/CD, Performance Monitoring, Splunk, JMX, Capistrano, PXE/TFTP, DHCP, DNS, Cobbler, Centos, MySQL/PostgreSQL, NoSQL, Cassandra, RabbitMQ, JMS, LDAP, SSO, Metrics, Graphite, Agile, RPC, Cloud, Clustering, Linux HA, HAProxy, Pacemaker/Corosync/Cman, SAN, GFS2, RAC, Volume Manager,
Nov. 2006 – Oct., 2010 Boomi Inc. Berwyn, PA,USA Senior Software Engineer	• Boomi On-Demand Business Integration Cloud Service • Cloud Platform/Infrastructure Design and Implementation for Boomi On Demand Production Environment • Desktop Accounting and ERP Enterprise Applications Integration (EAI) • Amazon EC2 and S3 SOA HA Environment • Setup Development, Build and Deployment Environment for QA, Staging and Production Environment • Research virtualization technology and implement Virtual Machines using VMware server and workstation • Java to C/C++, DLL, COM/COM+/DCOM/.NET Integration • SaaS system integration • SAP system integration Java, SOA, IaaS, PaaS, EAI, XML, XSD, JAXB, JavaScript, .NET, EC2, S3, EBS, REST, SOAP, WSDL, SaaS, SAP, BAPI, ABAP, MySQL/PostgreSQL, Maven, SVN, JVM, JVM GC Tuning, Apache Lucene, Apache Solr, VMWare, Cloud,

Jun., 2006 – Nov.,2006 Vantage learning Newtown, PA,USA Senior System Engineer Manager	• Large scale SOA e-learning system architecture design and blueprinting • Large Data Center Design and Administration • Sun SPARCStorage SSA Array 100 and Sybase SQL Server (10.x, 11.x) Data (DB schema, data and Stored-Procedure) Recovery • Ecommerce Open Source Software installation and Configuration • Resin Application Server Distributed Session Design and Setup
	Linux, Centos, SOA, Java, .NET, Network, VLAN, Router, Firewall, Load Balancer, SOAP/WSDL, MySQL/PostgreSQL, Struts, Disk Array, Fedora, Redhat Linux, LAMP, eCommerce, SQL,
May, 2002 – Jun.,22,2006 Computer Science, Texas Tech University Lubbock, Texas,USA Senior Programmer/Analyst and Unix System Administrator	• Unix network System and Software routine maintenance • Setup Veritas Netbackup System with RAID0/1/5 and Tape Library • Setup LDAP, Kerberos KDC and Cross-Realm system • Migrated email system from UNIX mail server to Windows Exchange Server • Secured Postfix SMTP server with SASL Authentication and Verification • Installed AIX 5.1 and configured NIM on IBM RS6000 (10x 7043-140 boxes and 1x 7024-E30 box) • Deployed Globus Toolkit3.2.1 (OGSA/OGSI) Grid Computing Environment on Solaris 9(3 boxes) and Linux (3 boxes) • Setup Jumpstart Environment for Solaris 9 and Solaris 10 • Designed and Implemented the Single Sign-On (SSO) for Unix and Windows Environments • The Migration from NIS+ to LDAP for Solaris
	JAVA, C/C++, GCC, GNU Make, perl, LADP, NFS, SSO, Kerberos, SSL, SMTP, Email Sever and Client, Solaris, AIX, Mac OSX, Linux, Windows Server, PAM, RAID, SCSI, Veritas, TFTP, RARP/ARP, Bootparamd, DHCP, Postfix, PostgreSQL, NIS/NIS+,
Aug.,2002 – Jun.,2006 Computer Science, Texas Tech University Lubbock, Texas,USA Master Student	• Master Thesis: A Service Wrapping and Provisioning Framework for SOA • GIS Project: A Yield Mapping and Prediction Information Delivery System • Independently finished practical course projects list
	Java, C/C++, GCC, Gnu Make, Linux, Linux Kernel, GIS, Ant, Struts, J2EE, UML2, MVC, Maven, PostGIS, PostgreSQL, Solaris,
Apr., 2000 – May, 2002 Technology Development Center, Sun Microsystem (China) Co.,Ltd. Beijing,P. R. China Senior Software Engineer	• Java Message Service(JMS) Prototype Application • Remote Learning Management Platform (www.ambow.com) • Bao Steel B2B Website (www.bsteel.com) • Project Management Tools
	Java, J2EE, JMS, EJB, Servlets, JDBC, Message-Driven Bean, LDAP, SQL, Weblogic, OOA/OOD, UML, MVC, Oracle DBMS, Java IDL, RMI, CORBA, JNDI, Solaris, CVS, Ant,
Aug.,1998 – Apr.,2000 IBM China Research and Development Lab Beijing,P. R. China Software Engineer	• Credit Card Internet Online Payment System • Jiro/Jini-based Storage Management System • Handwriting Recognizing System
	Java, J2EE, C/C++, Servlets, WebSphere, SSL, JDBC, Windows NT, SAN, POS, JNI, DB2,
Apr.,1996 – Jul.,1997 North China Institute (Taiji Computers Corporation) Beijing,P. R. China System/Network/Software Engineer	• System Engineer,IBM Service Center • Software Engineer, HUAXU Golden Card Co.Ltd. • Large scale Network System Design and Administration
	C/C++, Assembly, Network Design, Router, Switch, VLAN, AIX, Disk Array, SSA, RAID, Volume Manager, IC, RIP/OSPF, Ciso IOS,
Jun., 1991 – Sep.,1993 Yongji Electrical and Motor Factory ShanXi Province,P. R. China Mechanical Engineer	• CIMS and FMS CAPP/CAD/CAM manufacturing Automation application development
	C/C++, AutoCAD, CAD/CAM/CAPP, CIMS/FMS, DOS,

SELECTED PROJECTS

1. Dec.,2021 – Current, Senior Cloud Engineer II (Associate), McKinsey & Company , Cherry Hill, NJ (WFH),USA

Working on migrating/refactoring complicated EDA mainframe system to Cloud (AWS) serverless EDA system with EventBridge/Lambda/Step Function/SQS/SNS/CloudWatch/EKS .etc.

1.1 Migration/Refactoring of Mainframe (IBM MQ) EDA system to AWS by using EventBridge/Lambda/Step Function/CloudWatch/EKS/SQS/SNS...etc.

Design and Implement/Deploy the EKS Router to integrate AWS EventBridge with external MQ events

Technologies:

Java/SpringBoot, Terraform, Microservice, Java/JavaScript, EKS, AWS/EventBridge/CloudWatch/Lambda, XML/XSD/Json/Json Schema, DynamoDB, RDS/Oracle,

2. Dec.,2020 – Dec., 2021, Staff Engineer, Yotascale , Cherry Hill, NJ (WFH),USA

Cloud(AWS/Azure/GCP) Cost Analytics/Optimization/Management platform.

2.1 Lead Architect and Lead Implementation Engineer for Cloud(AWS/Azure/GCP) Cost Analytics/Optimization /Management platform

Design and Implement/Deploy the Cloud Cost Analytics/Optimization/Management platform to help different stakeholders to understand/manage the cloud cost from different perspectives (Lens: Organizational/Technology stack/Business domain...etc.).The platform ingests the raw cost data (CUR file..etc.) into data warehouse (ClickHouse) with kubeflow pipelines and allocates the cost data based on customizable allocation rules/trees. Metadata driven analytics widgets config and cost analytics Microservices with openAPI/Swagger spec are deployed in EKS with Istio and Auth0/JWT tokens for SSO authentication and external authorization for RBAC. Apollo GraphQL cluster is used to simplify the cache management and REST APIs aggregation. One page REACT front-end is developed to generate the cost analytics diagrams/graphs dynamically.

Technologies:

Cloud Cost Analytics/Optimization/Management, OpenAPI/Swagger, Microservice, Java/Kotlin/Python, Spring/SpringBoot/Flask, Kubernetes/EKS /AKS/GKS, Terraform/Terragrunt/Ansible, AWS/EC2/EBS/S3/VPC/Security/R53/IAM/Secrets Manager/Lambda, Service Mesh/Istio/Envoy, Helm/Chart, RabbitMQ, Azure/GCP, Junit/Test Containers, Docker/Minikube,

3. Mar.,2020 – Nov., 2021, Technical Infrastructure/Devops Architect, Gigster Inc. , Cherry Hill, NJ,USA

Cloud (AWS) Infrastructure architecture (CI/CD/Automation/Terraform/EKS) design and implementation

3.1 Images manipulation/detection software evaluation system

Lead Infrastructure /DevOps Architect/Implementation Engineer for Images manipulation/detection software evaluation system using Microservices (NodeJs) and EDA (RabbitMQ with PostgreSQL) within EKS cluster

Technologies:

Cloud Cost Analytics/Optimization/Management, OpenAPI/Swagger, Microservice, Java/Kotlin/Python, Spring/SpringBoot/Flask, Kubernetes/EKS /AKS/GKS, Terraform/Terragrunt/Ansible, AWS/EC2/EBS/S3/VPC/Security/R53/IAM/Secrets Manager/Lambda, Service Mesh/Istio/Envoy, Helm/Chart, RabbitMQ,

4. Jun.,2017 – Feb., 2020, VP,Lead Architect Engineer, Barclays , Wilmington, DE,USA

Cloud Adoption to migrate on-premises enterprise systems to public cloud (AWS). Delivery lead for Distributed System technology domain which includes Microservice, Event Driven Architecture (EDA), Message transports, integration (Batch/ETL) and (Event) Stream processing technologies.

4.1 Delivery Lead for strategic distributed system technology domain

Delivery lead for Distributed system technology domain which includes Microservice, Event Driven Architecture (EDA), Message transports, integration (Batch/ETL) and (Event) Stream processing technologies

Technologies:

Technology strategic Design and Stewardship, Distributed system, Domain Driven Design, Microservice, ACID/BASE, (Event) Stream /Batch Processing, Event Driven Architecture (EDA), Message Transports (Kafka, ActiveMQ, IBM MQ), Data Integration (ETL/ELT), Compensating transactions/Sagas, JMS/AMQP, Resilience and Performance, CQRS/SOA, CAP,

4.2 Group-wide Cloud Adoption for Contact Center, Merchant Services and Fraud

Cloud Assessment and consulting for cloud architecture for on-premises applications migration to AWS. Troubleshooting the cloud migration related issues

Technologies:

AWS, EC2, RDS, S3, ECS, VPC, ACL/Security Group/Routing table, ELB/NLB/ALB, Private Link, EBS,

5. May, 2014 – June, 2017, Lead Software Engineer, Oracle , Redwood, CA (WFH@NJ),USA

The Cloud Connectivity software engineering team is responsible to develop the Cloud Integration Connectivity SDK and Java Connectors which enable the SOA Suite and Oracle Integration Cloud Service (ICS) connectivity to any Oracle internal or external SaaS or On-Premises Enterprise Applications.

5.1 Docker Swarm/Compose/Stack CD/CI Test Environment for Cloud SDK/APIs/Connectors

Research and prototype the docker containers, compose and swarm technologies to help developer setup complex testing services stack and improve CI testing environments for the SDK/APIs and Java Connectors.

Technologies:

Docker, Docker Machine, Docker Compose, Docker Swarm, Docker Stack, Docker CLI (Rest APIs), WebLogic, Oracle Cloud VMs, AWS, JMS, NoSQL (Cassandra), DBMS (MySQL and PostgreSQL),

5.2 Cloud Enterprise Applications Integration(EAI) connectivity SDK/API plugin framework for Oracle SOA Suite

The Cloud Integration Connectivity SDK and Java Connectors are built on the top of the Connector Architecture in the Award winning Oracle SOA Suite. It provides the plugin connectivity to any SaaS and On-Premises Enterprise Applications for the Oracle Integration Cloud Service and SOA Suite.

Technologies:

Java, SOA, SOAP/REST, WSDL/JSON, XML, XML Schema, API, SDK, IaaS, RCU, JCA, XDK, JAXP, WSDL4J, DOM, WS-Security, Jersey REST/JSON, MVC, ADF, NLS/L10, Oracle DB, Jira, Agile, ADE, GIT, Maven, Junit, XMLUnit, SoapUI, OSC, ERP, HCM, RightNow,

5.3 Oracle Fusion Applications (FA) Business Event/Message-Driven Connector Framework

The business event connector framework provides the abilities to auto-discover the FA (OSS/ERP/HCM) business events (including custom events), set event filtering conditions, allow the end user to subscribe to interesting business event and expose the end user's SOAP web service as the event listener through the ICS web/JDeveloper design time UI. This event connector framework enables the Fire-and-Forget/ Asynchronous Response and Publish/Subscribe integration pattern for Business Event/Message Integration

Technologies:

Java, FA, OSC/ERP/HCM, WSDL/SOAP, REST, JSON, XSI, XML Schema, XDK, JAXP, WSDL4J, DOM, WS-Security, Jersey REST/JSON, MVC, ADF, ADE, Maven, Junit, XMLUnit, SoapUI,

6. Oct. 2010 – Apr. 2014, Senior Cloud DevOps/Infrastructure Automation Engineer, Comcast T&P , Philadelphia, PA,USA

As a senior engineer in BITT(Build Integration Test Team), provided DevOps support for multiple development teams and supporting the developers for SDLC infrastructure system design/implementation, and Build/Release/Deployment/SCM/CI automation., designed, developed and implemented the AutoBuild system, OpenSource performance monitoring system for the SDLC systems, two steps bare-metal Linux bootstrap system for HP servers, Linux HA Clusters for SCM/Maven/CI/Crowd servers, Ruby/Rails software system maintenance and improvement. Managed the SCM(Gerrit/Subversion), CI(Bamboo/Jenkins) and maven repo (Nexus and Artifactory) system. Designed and developed automation scripts (bash/ruby) to improve work efficiency. Led and Trained junior team members.

6.1 AutoBuild System for Build/Release more than 200 Microservice projects with GIT/SVN/Maven

Automate the branching/tagging process and improve the work efficiency from 3 days to 1 day for more than 200 projects. Using maven versions plugin and maven repo REST API to automatically detect the latest snapshot and released dependency version and update pom.xml automatically. The cut tag script can cut a bug-fixing tag for the bug-fixing patch within 5 minutes.

Technologies:

Ruby/Rails, gems, Bash, Maven, GIT, SVN, Bamboo/Jira/Confluence CLI, Nexus/Artifactory REST/XML, SDLC, CI/CD, Gerrit/Subversion, MySQL/PostgreSQL, Crowd/OpenID, Capistrano,

6.2 Configuration management and deployment automation with Puppet and Capistrano

Configuration management and deployment automation system by using puppet for linux os level configuration and capistrano for application level deployment automation

Technologies:

Puppet, Capistrano, Ruby, bash, CI/CD, GIT, Maven, Perl, Python,

6.3 Continuous Integration (CI) and Continuous Deployment (CD) with Bamboo and Jenkins

Automation and Integration of the CI/CD clustering environments for more than 200 Microservice with build/runtime dependencies using Bamboo and Jenkins

Technologies:

CI/CD, Bamboo, Jenkins, Ruby, Bash, Python, GIT, Maven, Jira/Confluence,

6.4 Performance Monitoring System for Jetty, JVM, Linux and Splunk

The performance monitoring system collects JMX, splunk and OS metrics and display stats graphs for all different kinds of metrics for JVM heaps, threads and OS CPU/Disk/Memory usage and IO throughput. Using open source projects and modify ruby/scala/java/python tools to integrate all different projects into one performance monitoring system, the system helps to detect the performance bottlenecks.

Technologies:

Metrics, graphite, jmx, splunk, Ruby, splunk-ruby-SDK, performance monitoring, UDP, Capistrano, god, bash, gdash, graphite, statsd, metricsd,

6.5 Secured FTP server with virtual users and admins users with complex access permissions

Setup vsftp with virtual ftp only users and sftp servers with chrooted admin users to get complex access control for different type of users

Technologies:

ftp, sftp, chroot, bash, linux, user/group unix permission, sshd,

6.6 BOM (Bill Of Materials) Administration and Management Application (BAMA) Enhancement and maintenance

Implemented with Ruby on Rails 3, Delayed_jobs, Ruby Websocket server, mySQL as the DB server, Passengers(mod_rails) and Apache HTTPD as LB, an internal BOM Administration and Management Application provides deployment automation and Management for more than 200 Microservice for different environments (Dev/QA/Staging/Prod).The Enhancement feature enables the user to customize the list of Microservice to deploy to different systems with matched Capistrano versions and BOM versions.

Technologies:

Ruby, RoR 3, Rails 3, ActiveRecord, ActionView, ActionController, ActionMailer, Websocket, linux, JSON, Capistrano,

6.7 Two Steps bare-metal Linux bootstrap system for HP Servers

This system bootstrap the bare-metal HP box to automatically install and configure CentOS 6. First, it boots the HP bare-metal box to the HP-Toolkit via tftp/pxe to setup BIOS configuration, collect the mac address, add DHCP host section with static ip and add new cobbler system using customized cobbler xmlrpc bash client. Second, it boots the bare-metal box to get the static DHCP IP and tftp/pxe to the cobbler profile to install and configure CentOS 6 automatically.

Technologies:

TFTP, PXE, DHCP, DNS, cobbler, RPC, xml, bash, ILO 3, IPMI, LVM, HP Servers, CentOS 6,

6.8 Linux HA for Git Repo(Gerrit) and Maven Repo (Artifactory/Nexus) Servers

The two nodes Linux HA cluster is for Git repo (Gerrit) / Maven Repo(Artifactory and Nexus) master/slave failover system. Using pacemaker/corosync/cman or keepalived/haproxy, it setup Gerrit, artifactory/Nexus master/slave automatically failover cluster with PostgreSQL/RAC and GFS2 over SAN / DRBD as backend DBMS and distributed file system.

Technologies:

Pacemaker, Corosync, Cman, Keepalived, HAProxy, GFS2, DRBD, SAN, Gerrit, Artifactory, Nexus, PostgreSQL, OpenID, LDAP, Crowd,

6.9 Automation scripts and tools for DevOps

Develop and use scripts (Ruby/Bash) and tools to automate the build/release/deploy process to improve the work efficiency.

Technologies:

Ruby, Bash, Python, Capistrano, Jira/Rally, Bamboo/Confluence, Git/Gerrit, puppet, Nexus replication, REST/XML,

6.10 CoreOS and Docker prototyping and testing

Using docker and CoreOS to dynamically create development and integration test environment on demand for each developer by launch a set of docker LXC with haproxy as the internal load balancer

Technologies:

CoreOS, Docker, Ruby, REST, LXC, cobbler, pxe, tftboot, DHCP,

6.11 SDLC process management and junior team member training and mentoring

Design and Manage the SDLC process to build/release/deploy more than 200 projects to maven repo/CI/Integration Test environments. Maintain the maven repo servers(Nexus and artifactory clusters), CI (Bamboo), SCM (Subversion and Git/Gerrit) and clustering integration environments. Identify and troubleshooting the build/release/deploy issues. Training, guiding and mentoring junior team member.

Technologies:

SDLC, SCM, Git/Subversion, GIT Hooks, Ruby, Bash, Confluence, Jira, Agile,

7. Nov. 2006 – Oct., 2010, Senior Software Engineer, Boomi Inc. , Berwyn, PA,USA

As the market and technology leader in on-demand integration, Boomi is the industry's first and leading integration platform-as-a-service and a two-time SIIA CODiE award winner - Best Application Integration Solution (2010) and Best On-Demand Platform (2009).

7.1 Boomi On-Demand Business Integration Cloud Service

A Large On-Demand integration service-oriented (SOA) system helps to visually design the business integration process using Rich Internet Application (RIA), automatically deploy to distributed run-time engines based on REST/XML style. The run-time engine named "atom" contains all the components required to execute an integration process from end-to-end including connectors, transformation rules, decision handling and processing logic. The software is delivered as a SaaS system based on web 2.0 technologies with zero installation and zero coding.

Technologies:

Java, XML, XSD, JAXB, REST, HTTP, Hibernate, openlaszlo, SSL, PKI, com4j, .NET 3.0/2.0/1.1, COM/DCOM, Linux, QXML, eConnect, servlet, Jetty, IDEA, Maven, subversion, Confluence, Jira, MySQL, RAID, SQL server, QuickBooks 2005, Sage MAS 90/200/500, MS Dynamics GP 9.0/10.0 /2010, SOA,

7.2 Cloud Platform/Infrastructure Design and Implementation for Boomi On Demand Production Environment

Distributed clustering production environment with ten 64-bits Redhat Enterprise Linux servers, Clustering MySQL DB servers, SAN and NFS is hosted in Rackspace and OpSource.

Technologies:

64-bits Redhat Enterprise Linux, MySQL clustering, Apache, PHP, Jetty, 64 bits Java, Apache Lucene/ Solr, Openlaszlo, distributed cache, distributed and parallel algorithm, ehcache, comet, continuation, DWR, JVM tuning, jmap, jstat, Sun JVM Heap, GC, vmstat,

7.3 Desktop Accounting and ERP Enterprise Applications Integration (EAI)

Solutions integrating with some popular desktop accounting software (QuickBooks, MS Dynamics Great Plains, Sage Peachtree and Sage Simply Accounting.), and ERP(SAP, Oracle e-Business, PeopleSoft, JD Edwards and Siebel, MS Dynamics Navision, MS Dynamics CRM On-premise 4.0 and online) are developed to import and export business data using their related popular SDKs and APIs.

Technologies:

QXML RequestProcessor, Microsoft eConnect 9.x/10.x/11, Librados, SAP R3, BAPI, REST/XML, COM/COM+/DCOM, .NET, MS Dynamics Navision, MS Dynamics CRM, SOAP/WSDL, WCF, XMLPort, C/AL, CodeUnits,

7.4 Amazon EC2 and S3 SOA HA Environment

Three EC2 instances including two web instances as a cluster and one DB (MySQL 5.x) instance, which is installed and configured with Apache open source search server solr and Apache httpd load balancer, are configured in Amazon web service virtual environment.

Technologies:

Fedora 6/7/8, AWS, EC2, S3, Jetty 6.x/7.x, MySQL 5.x, Apache httpd 2.x and Solr 1.3.x/1.4.x, NFS, shell script(bash and sh), ehCache, cron,

7.5 Setup Development, Build and Deployment Environment for QA, Staging and Production Environment

Setup java, maven, subversion development, TeamCity build environment and Apache ant release xml, shell scripts and PHP scripts to maintain the user accounts.

Technologies:

Maven 2.x, subversion, TeamCity, PHP 5.x, Apache Http server 2.3, Apache Ant, MySQL5.x, Jetty 6.x, Openlaszlo 3.4, Apache Solr, Fedora 8,

7.6 Research virtualization technology and implement Virtual Machines using VMware server and workstation

Researching and comparing current popular virtualization and paravirtualization technologies like VMware, Xen, Microsoft Virtual PC and implementing Virtualization Environments using VMware Workstation 6.x and Server 1.x. Performance diagnostics and tuning in Linux and VMware Environment.

Technologies:

Virtualization, paravirtualization, VMware Workstation 6.2 and VMware Server 1.4, Xen, Microsoft Virtual PC, Fedora 8, RAID, SMP, Performance tuning, Subversion, Maven, PHP, MySQL,

7.7 Java to C/C++, DLL, COM/COM+/DCOM/.NET Integration

Using JNI, JNA, Com4j, ikvm, java application can be seamlessly integrated with c/c++ win32 DLLs, COM/COM+/DCOM Objects and .NET assemblies.

Technologies:

Java, .NET 3.0/2.0/1.1, COM/DCOM, Com4j, iKVM, C#, VB.NET, JNI, JNA, Win32 DLL, COMAdmin,

7.8 SaaS system integration

Design, development and implement integration solutions for SaaS systems (Parature, AutoTask and Zuora) based on REST/XML and web services (SOAP/WSDL) technologies.

Technologies:

REST/XML, SOAP/WSDL, namespace, DOM, soapUI,

7.9 SAP system integration

Integrate with SAP NetWeaver and SAP R3 ECC 6.x via web services and librados (sapjco)

Technologies:

SAP R3, NetWeaver, Web Services, SOAP, WSDL, wsdl4j, SAP-JCO, RFC, BAPI, ABAP, Java,

8. Jun., 2006 – Nov., 2006, Senior System Engineer Manager, Vantage learning , Newtown, PA, USA

As the leader in cost-effective, high volume, secure, scalable online assessment and instructional programs for K-12 and higher education, Vantage Learning leverage technology such as artificial intelligence, natural language understanding, and web-based learning objects. Vantage Learning has received accolades ranging from the prestigious CODIE Award for best instructional technology.

8.1 Large scale SOA e-learning system architecture design and blueprinting

Designed and blueprinted large scale SOA e-learning system based on Java/.NET with PostgreSQL and MySQL DB clusters

Technologies:

SOA, E-Learning, Struts, J2EE, .NET 2.0/3.0, Resin, SOAP, WSDL, UDDI, XML, PostgreSQL, MySQL,

8.2 Large Data Center Design and Administration

More than 100 Unix/Linux/Windows server machines and enterprise network devices (three sets of BIP load balancers, switch and routers) data center with more 30 mission-critical applications serving millions of Internet connections.

Technologies:

Solaris 5,7, Linux(Redhat,Fedora), F5 Big-IP load balancers, Switch, Router, VLAN, UPS, RAID 1,0,5, Windows Server, Terminal server, VNC, SPARCStorage, Tape Drive, Veritas, Apache, Resin, PostgreSQL,

8.3 Sun SPARCStorage SSA Array 100 and Sybase SQL Server (10.x, 11.x) Data (DB schema, data and Stored-Procedure) Recovery

Recovered important patent data and technologies from an old dead Sun SPARCStorage SSA Array with Sybase Server

Technologies:

Solaris 5,7, Linux(Redhat, Fedora), Sybase SQL Server 10.x, Sybase ASE 11.x, 12.x 15.x, SSH, SPARCStorage SSA Array 100, Tape Drive,

8.4 Ecommerce Open Source Software installation and Configuration

Installed and Configured the osCommerce with the x-cart and Sales-n-Stats modules in PHP and MySQL environment

Technologies:

Linux(Redhat, Fedora), PHP5.x, MySQL 5.x, Apache 2.x,

8.5 Resin Application Server Distributed Session Design and Setup

Installed and Configured Resin Application Server with distributed session and http load balancer

Technologies:

Linux(Redhat, Fedora), resin 3.0, Java, Servlet, distributed session, cluster, load balancer (software and hardware),

9. May, 2002 – Jun., 22, 2006, Senior Programmer/Analyst and Unix System Administrator, Computer Science, Texas Tech University, Lubbock, Texas, USA

System Administrator for networked Unix(Solaris, AIX, Mac OSX and Linux) and windows servers with SSO for Kerberos and LDAP, Veritas Netbackup for Disk Array with RAID 0/1/5, Migration from NIS/NIS+ to LDAP and Postfix/Sendmail configuration for the research and teaching lab for computer science department.

9.1 Unix network System and Software routine maintenance

Daily maintenance and system administration for networking Solaris, AIX and Mac OSX

Technologies:

Solaris 8,9,10, Linux, AIX, NFS, NIS/NIS+, LDAP, SMTP, IMAP, POP3, WebMail, SSH, Samba, Kerberos, Netgroup, Jumpstart, User Accounts, Quota, License server, GNU Make, GCC, Java, RAID, Tape Library, Veritas, Apache, MySQL, Sun Ray Server, PAM, Security, Syslog,

9.2 Setup Veritas Netbackup System with RAID0/1/5 and Tape Library

Setup, Administration and Backup for large Disk Array (RAID 0/1/5) Storage and virtual File system with Volume Manager

Technologies:

Veritas NetBackup Enterprise Server 5.1, Veritas NetBackup Client, RAID 0,1,5, SCSI, Volume Manager, VFS, Tape Library,

9.3 Setup LDAP, Kerberos KDC and Cross-Realm system

Designed and Implemented the Single Sign On (SSO) with setup LDAP server, Kerberos KDC and Cross-Realm System authentication

Technologies:

SASL-2.1, GSSAPI, SEAM 1.2, SunOne Directory Server 5.2, NIS+, LDAP V3, PAM, Solaris 9, Windows Server 2003, Active Directory,

9.4 Migrated email system from UNIX mail server to Windows Exchange Server

Setup, Administration for SMTP servers in Solaris and Windows and Configured all different kinds of email clients for all different systems with POP3/IMAP with SSL authentication

Technologies:

Sendmail, Postfix, Unix Mail Client, Outlook, Solaris 9, Linux, Windows Server, Exchange Server, IMAP/S, POP3/S, SSL,

9.5 Secured Postfix SMTP server with SASL Authentication and Verification

Setup and configured Postfix SMTP server with SASL Authentication and Verification

Technologies:

Openssl-0.9.7, SASL-2.1, Postfix 2.4, SquirrelMail, pine, Solaris 9, Linux,

9.6 Installed AIX 5.1 and configured NIM on IBM RS6000 (10x 7043-140 boxes and 1x 7024-E30 box)

Setup and configured IBM AIX for RS6000 with NIM and NFS server

Technologies:

AIX 5.1, NIM, NFS, NIS/NIS+, SORCER/Jini/Rio,

9.7 Deployed Globus Toolkit3.2.1 (OGSA/OGSI) Grid Computing Environment on Solaris 9(3 boxes) and Linux (3 boxes)

Setup and Configured Globus Toolkit Grid Computing Environment with SSL certificates

Technologies:

Unix (Solaris 9, Redhat Linux 9), NFS, NIS/NIS+, Perl, PKI, X509, GNU Tools, Globus Toolkit 3.2.1, OGSA/OGSI, GSI,

9.8 Setup Jumpstart Environment for Solaris 9 and Solaris 10

Designed and Implemented the Jumpstart Environment with one installation server, one profile server and two boot servers for two different IP subnets to install and configure network, NIS+ , NFS for more than 50 Sun client boxes. Developed Perl scripts to maintain NIS+ tables and tftp boot parameters

Technologies:

Unix(Solaris 9), Jumpstart, TFTP, Bootparamd, NIS+, Perl, Security, snoop, RARP/ARP,

9.9 Designed and Implemented the Single Sign-On (SSO) for Unix and Windows Environments

Integrate the UNIX and Windows environment in a Single sign-on environment using LDAP, Kerberos, AFS, Samba technologies

Technologies:

Unix(Debian Linux, Solaris 8 and 9), LDAP(OpenLDAP2.1.2, SunONE Directory Server 5.2 , MS Active Directory), AFS (OpenAFS 1.2.10), Kerberos (SEAM 1.2, MIT Kerberos V 1.3.1). PAM, GSSAPI, SSL, SASL, SSH, C++, GNU make, GCC, Visual.NET, Perl,

9.10 The Migration from NIS+ to LDAP for Solaris

Migrated the NIS+ services in Solaris 8 to LDAP (iPlanet Directory Server 5.1.) in Solaris 9. Installed, configured and implemented the test environment and designed LDAP schema to satisfy department's requirements in the directory services. Developed Perl scripts running on Solaris 9 to fulfill user managements.

Technologies:

Unix(Solaris 8 and 9), LDAP(SunONE Directory Server 5.1), Kerberos (SEAM 1.2). NIS/NIS+, SSL, SSH, Perl, GNU make , GCC,

10. Aug.,2002 – Jun.,2006, Master Student, Computer Science, Texas Tech University , Lubbock, Texas,USA

Studied as Master Student in the CS department while working as part time as System Administration for the Teaching and Research labs for Unix/Linux and Windows environments

10.1 Master Thesis: A Service Wrapping and Provisioning Framework for SOA

Based on Jini/Rio technologies, this framework automatically generates the service wrapper codes (service dynamic smart proxy, bean wrapper and provisioning descriptions in xml format ...etc.) for legacy code and autonomic provision it as a service provider (JSB - Jini Service Bean) into SOA grid environment. A GIS based Yield Tracker system (developed in Basic language) is wrapped as service provider and provisioned into multiple Rio cyber nodes.

Technologies:

Jini 2.0, RIO, JERI, Globus Toolkits 3.x, J2EE, Web Service, OpenGIS, PostGIS, JDBC, GeoServer, Struts, Ant, Tomcat, Servlet, JSP, UML2,

10.2 GIS Project: A Yield Mapping and Prediction Information Delivery System

By using a OpenGIS project Geoserver and GIS DBMS (PostgreSQL/ PostGIS) , the system displays a real web GIS map to the farmer. The farmer selects a particular field and other related planting information (planting date, irrigation or not, crop type) from the web map. The system uses an agriculture model (developed in Basic Language) and other information collected from satellites system in real-time to dynamically generate a web GIS based yield prediction map.

Technologies:

OpenGIS, PostGIS, PostgreSQL, JDBC, GeoServer, GeoTools, Struts, Ant, Apache Maven, Servlet, JSP, MVC,

10.3 Independently finished practical course projects list

Technologies:

GCC Compiler, Solaris, Linux Kernel 2.4, UML 2, Rational Rose, Java, SORCER, SOA, Jini, JavaSpace, JBoss, J2EE, VFS, Java Card platform, MIDP, MIDlets, Servlets, OCF, CLDC, Ant,

11. Apr., 2000 – May, 2002, Senior Software Engineer, Technology Development Center, Sun Microsystems (China) Co., Ltd. , Beijing, P. R. China

As Senior software engineer, provided consulting services for Java/J2EE technologies for local companies. Helped design and architecture the Java/J2EE systems around Sun's software products like iPlanet Application Server, Directory Server, Message Queue Server

11.1 Java Message Service(JMS) Prototype Application

The prototype proves the idea of using JMS to develop a software system to implement tax documents approval process in Tax Bureau of China. Two computer nodes were deployed to emulate city node and province/state node. Each node includes JSPs as input interface to human user, Servlets as messages sender to JMS message queue, Message-Driven Beans (MDBs) act as JMS message listener to message queue and also include JDBC connectors to local DBMS. This prototype fulfills a typical JMS system which includes the message queue server and application server and DBMS.

Technologies:

iPlanet Application 6.0, iPlanet Directory Server, iPlanet Message Queue Server, Oracle 8.x, Solaris, JSP, Servlet, EJB, JDBC, JNDI, LDAP, SQL, JMS, Message-Driven Bean, JMS, J2EE,

11.2 Remote Learning Management Platform (www.ambow.com)

Based on J2EE technology, an e-learning platform includes course management, certificate management, and platform management subsystems as well as other useful utilities. The development of the platform was followed the RUP engineering process. UML, object-oriented analysis (OOA) and object-oriented design (OOD) technologies were used during the whole modeling process.

Technologies:

Weblogic 6.0, Oracle 8.x, Windows 2000, Solaris, Rational Rose 2000, Microsoft Visual SourceSafe 6.0, JSP, Servlet, EJB, JDBC, JNDI, LDAP, SQL, Security, UML, OOA, OOD, MVC,

11.3 Bao Steel B2B Website (www.bsteel.com)

Based on the blueprint of the J2EE's sample program (Pet store) which implemented the MVC pattern, the first large-scale B2B website in China consists of more than ten modules such as User Manager, DB Manager, Auction, Products, Ordering, Bargaining, Communication, ...etc. and more than 100 tables.

Technologies:

iPlanet Application Server 6.0, iPlanet Directory Server 4.1, iPlanet Web Server 4.1, Oracle 8.1.5, Solaris 2.7, Window NT 4.0-Sp6, Jakarta Ant 1.1, CVS, JSP, Servlets, EJB, JDBC, JNDI, LDAP, SQL, Security,

11.4 Project Management Tools

Run on iPlanet Application Server (iAS) 6.0 platform, the tools provide the web-based functionalities such as project management, user management, document management and resources management. The user management module is LDAP-based authentication and session-based authorization. A CORBA (JavaIDL) server provides the SCC and file accessing in local machine for EJBs (CORBA client) in iAS6.0. These tools include 30 Servlets, 20 JSPs, 10 EJBs and some other helper classes

Technologies:

iPlanet Application Server 6.0, iPlanet Directory Server 4.1, iPlanet Web Server 4.1, Oracle 8.1.5, Solaris 2.7, JSP, Servlet, EJB, JDBC, Java IDL, CORBA, JNDI, LDAP, SQL, Security,

12. Aug.,1998 – Apr.,2000, Software Engineer, IBM China Research and Development Lab , Beijing, P. R. China

As software engineer, designed and implemented Java/J2EE systems around IBM software products like IBM WebSphere ...etc.

12.1 Credit Card Internet Online Payment System

Based on the existing POS payment system supported by ICBC(Industrial and Commercial Bank of China), a credit card online payment system was developed and deployed using IBM Payment Server, Net.Commerce (previous version of IBM WebSphere Commerce Suite), WebSphere Application server. Payment Server connected to ICBC POS front server via encrypted communication by using a serial (COM port) interface software module which emulates the typical POS credit card operation for the internet online user.

Technologies:

WebSphere Application Server, Net.Commerce, Payment Server, DB2, WinNT Server, JSP, Servlet, Security, SSL, JDBC, C, Socket and Serial Communication Programming, javax.comm, SDLC,

12.2 Jiro/Jini-based Storage Management System

Based on Jiro/Jini technology, developed a distributed management system for SAN.

Technologies:

Apache1.3, JDK1.2.x, Solaris 2.7, JSP, Servlet, Jiro/jini, RMI, SAN,

12.3 Handwriting Recognizing System

Using Java JNI, MFC, Lotus Domino, LS:DO, RDO:ODBC and DB2, developed a handwriting recognition testing, research environment for IBM ThinkScribe. It includes a web-based client (servlets communicate with ThinkScribe Driver via JNI), DB accessing (LS:DO and RDO:ODBC) and stand-alone application client (MFC)

Technologies:

Lotus Domino R5, Visual Basic 6.0, DB2, WinNT-sp5, Lotus Notes, MS-IE, JSP, Servlet, Lotus Script, LS:DO, RDO:ODBC, Visual Basic, MFC,

13. Apr.,1996 – Jul.,1997, System/Network/Software Engineer, North China Institute (Taiji Computers Corporation) , Beijing, P. R. China

Worked for different teams as System/Network/Software Engineer. Provided production supporting services for IBM RS6000 and Disk Array products. Developed IC Card Application system. Designed Large scale network system with Cisco Routers and Switches.

13.1 System Engineer, IBM Service Center

This service center is the joint department of IBM and Taiji Computers Company. It provides professional technical supports for IBM RS6000 and IBM Disk Array Products.

Technologies:

IBM RS6000 (SP2, R50, ..., 43P), AIX, IBM Disk Array, AIX, TCP/IP, HACMP, RAID0,1,5, SCSI, JFS, SSA, Volume Manager, Logical/Physical Partition,

13.2 Software Engineer, HUAXU Golden Card Co.Ltd.

Developed IC (Integrated Circuit) Card related hardware and IC Card application system

Technologies:

ATMEL Chips, Motorola, Memory/Security/CPU Card. M51 (Single Chip Computer). Visual C++, Visual Basic, Borland C++, Power Builder, Windows NT, C++, DLL, Lib, COS (Chip OS), DES, Assembly Language, FoxPro,

13.3 Large scale Network System Design and Administration

Designed large-scale network system plan and implemented the network system (hardware installation, Data link layer and IP layer configuration) with fiber network products (Cisco, IBM ATM and 3Com)

Technologies:

IBM 8265/8285/8274, Cisco Router 7500, Cisco 2511, ATM, ELAN, VLAN, HFC, RIP, OSPF, Cisco IOS, X.25, Frame-Relay, ISDN, Ethernet,

14. Jun., 1991 – Sep., 1993, Mechanical Engineer, Yongji Electrical and Motor Factory, ShanXi Province, P. R. China

Designed Locomotive engine electric motor/generator cases and automated with AutoCAD, CAPP and CAM with FMS/CIMS

14.1 CIMS and FMS CAPP/CAD/CAM manufacturing Automation application development

Locomotive's motor and generator related mechanical products design and manufacturing processing. Programming on FMS/CIM manufacture centers from GE (General Electric)

Technologies:

AutoCAD, C, C++, DOS, Z80(Single Chip Computer), Borland C++,

HOBBIES PROJECTS:

H.1 Personalized JSON Schema Resume

Customized the open source project dynamic-json-resume using JSON format and mustache templates with Node.js and CSS3. GitHub url: <https://github.com/rongyj/dynamic-json-resume/tree/rongyj>

Technologies:

JavaScript, HTML5, CSS3, Node.js, JSON, mustache, Puppeteer,